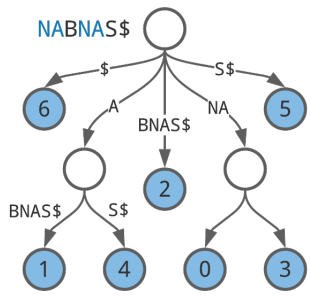
9D Find the Longest Repeat in a String

Longest Repeat Problem
Find the longest repeat in a string.

Input: A string *Text*.
Output: A longest substring of *Text* that appears in *Text* more than once.



Formatting

Input: A string *Text*.
Output: The longest substring that appears more than once in *Text*.

Constraints

- The length of *Text* will be between 1 and 10^3.

Test Cases

Case 1

Description: The sample dataset is not actually run on your code.

Input:
`panamabananas`

Output:
`ana`

Case 2

Description: The longest repeating sequence in *Text* overlaps with another instance of itself.

Input:
`GAGAGAG`

Output:
`GAGAG`

Case 3

Description: Multiple repeats occur with the same frequency in *Text* and are the same length.

Input:
`AGAGCTCT`

Output:
`AG` *or* `CT` (*you will not be penalized for having one over the other, but make sure you only output one*).

Case 4

Description: Repeats that occur at the beginning and end of *Text*.

Input:
`AAGCTGAA`

Output:
`AA`

Case 5

Description: There are no repeats in *Text*.

Input:

```
ABCDEFG
```

Output:

Case 6

Description: A larger dataset of the same size as that provided by the randomized autograder.